

About My Self

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I'm excited to share insights into the foundational concepts of Artificial Intelligence.



AI Basic: Foundational Concepts

At its core, Artificial Intelligence seeks to enable machines to perform tasks that typically require human intelligence. This involves a range of abilities, from perception and understanding to reasoning and learning.

Perception
Interpreting sensory data like images, sounds, and text to understand the environment.

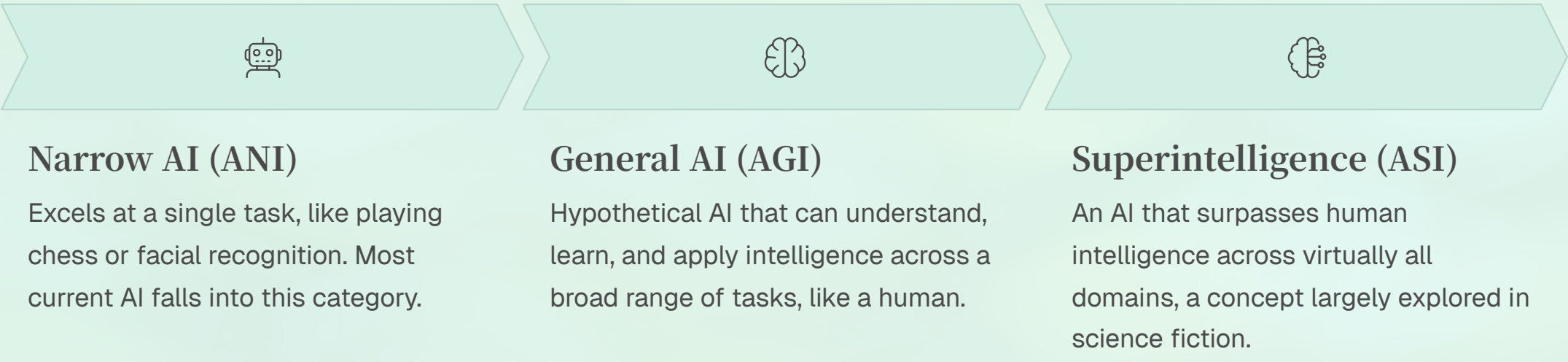
Reasoning
Solving problems, making decisions, and drawing conclusions based on available information.

Learning
Acquiring new knowledge and improving performance through experience and data.

Action
Executing tasks and interacting with the world, either physically or digitally.

The AI Spectrum: From Narrow to General Intelligence

AI can be categorized by its scope and capabilities, from systems designed for specific tasks to hypothetical human-level intelligence.



Artificial Intelligence: A 4-Month Academic Journey

From Theory to Game Theory

Defining Intelligence Science of creating agents that perceive, reason, and act.	Agent Evolution From Simple Reflex to complex Learning Agents.	Adversarial Search Implementing Minimax and Alpha-Beta Pruning for games.
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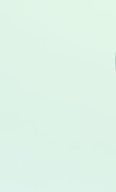
The Architecture of Intelligence: From Reaction to Learning



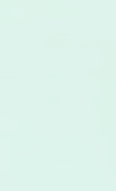
Simple Reflex
Reacts to current perception.



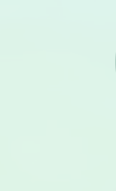
Model-Based
Maintains internal state.



Goal-Based
Plans ahead to reach a goal.



Utility-Based
Optimizes outcomes for maximum utility.



Learning Agent
Improves through experience.



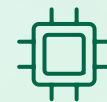
Agent-Environment Interaction: Sensing, Thinking, Acting

Intelligent agents are defined by their ability to interact with and navigate their environment. This fundamental relationship drives the design and capabilities of any AI system.



Sensing the World

Agents perceive their environment through sensors, collecting raw data that informs their internal state and understanding.



Processing & Decision

Based on perceptions, agents process information, build internal models, and make decisions to achieve their goals.



Acting on the Environment

Actions are executed through effectors, allowing agents to manipulate or change their environment in pursuit of objectives.



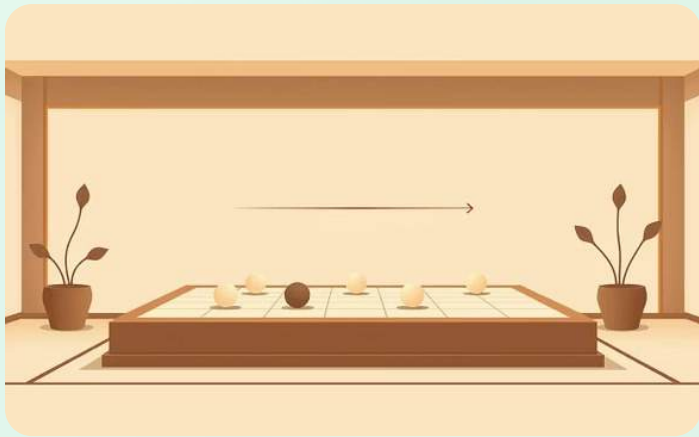
Diverse Environments

The characteristics of the environment (observable, stochastic, dynamic) critically shape the agent's architecture and complexity.

Understanding these interaction loops is crucial for developing robust and adaptable AI solutions that can thrive in various real-world scenarios.

When Intelligence Competes: Adversarial Search in Games

We implemented Minimax to model perfect play in zero-sum games and used Alpha-Beta Pruning to optimize search depth.



Tic-Tac-Toe

Minimax agent guarantees at least a draw by evaluating all moves.



Chess

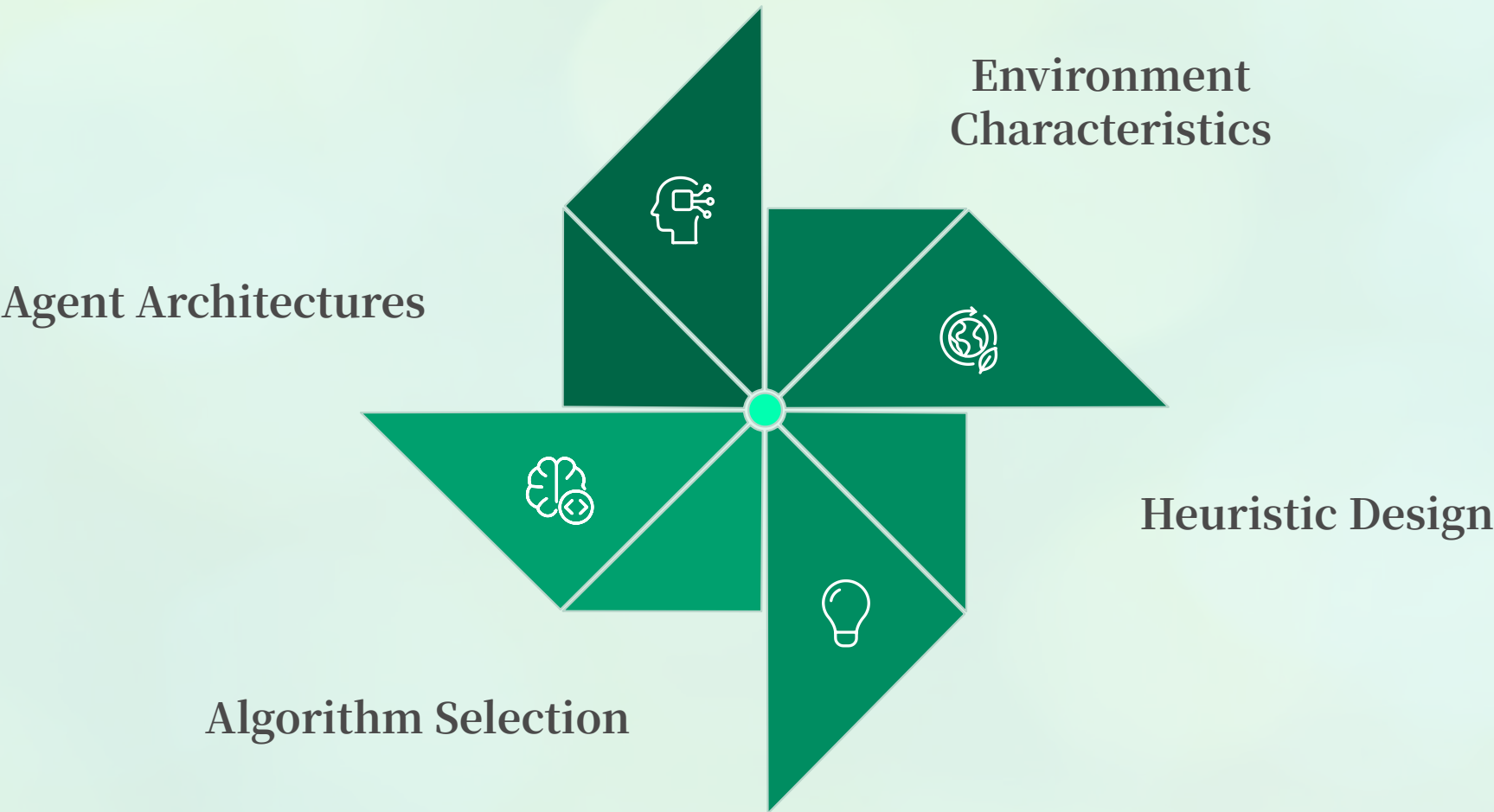
Alpha-Beta pruning allows the AI to search deeper by eliminating irrelevant branches.



Strategic Nim

Applied combinatorial game theory to always find winning positions.

Connecting the Dots: From Agents to Applications



Types of Search Algorithms

Search algorithms are fundamental to AI, allowing systems to explore a space of possibilities to find a desired goal or solution. They are broadly categorized based on how they use information to guide their exploration.



Uninformed Search (Blind Search)

These algorithms explore the search space systematically without any domain-specific knowledge. They guarantee finding a solution if one exists but can be computationally expensive.

Examples: Breadth-First Search (BFS), Depth-First Search (DFS).



Informed Search (Heuristic Search)

Informed search uses problem-specific knowledge, often in the form of a heuristic function, to guide the search and prioritize more promising paths. This significantly improves efficiency.

Examples: A Search, Greedy Best-First Search.*



Local Search

Unlike global search, local search algorithms operate on a single candidate solution, iteratively improving it by making small changes. They are often used for optimization problems where finding an optimal state is the goal.

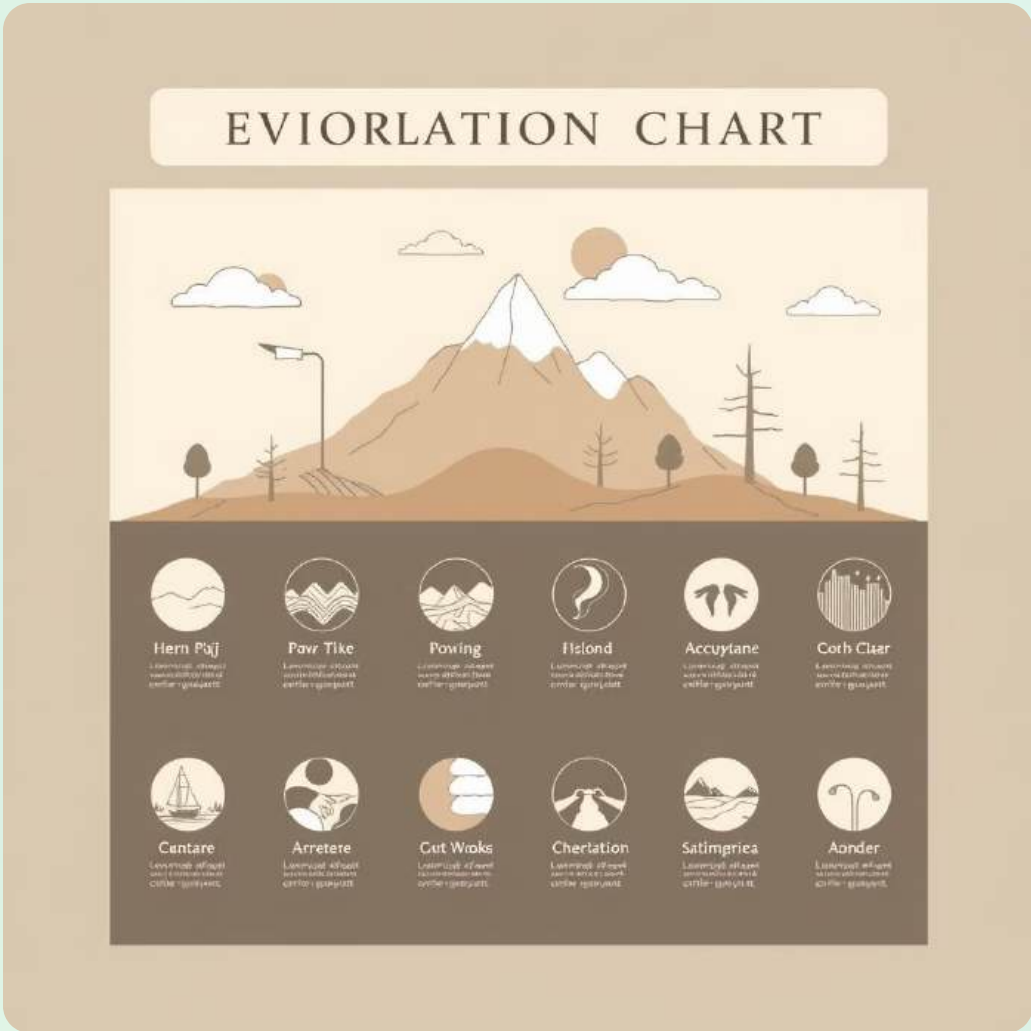
Examples: Hill Climbing, Simulated Annealing.

Problem Solving: Search and Optimization

Where AI Lives: Types of Environments

Environment characteristics influence algorithm choice:

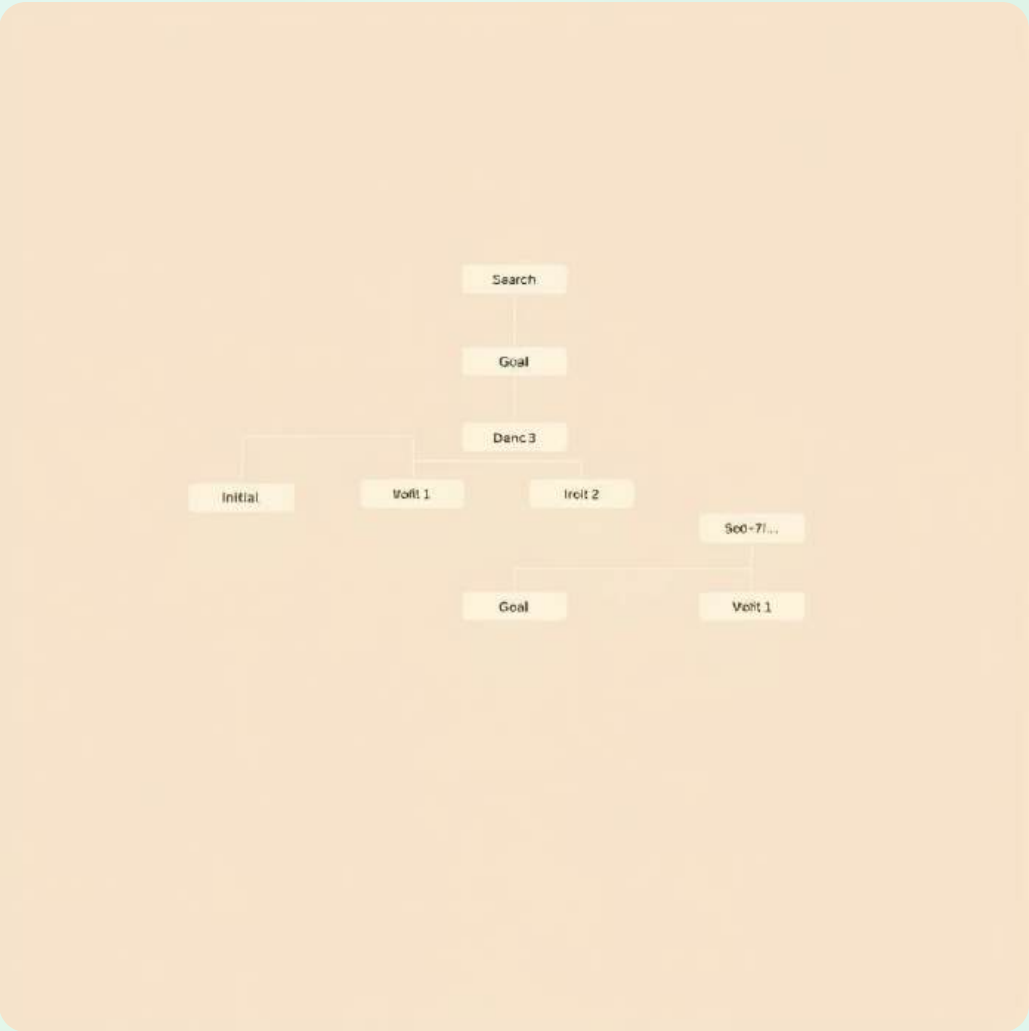
- Fully vs. Partially Observable
- Deterministic vs. Stochastic
- Static vs. Dynamic
- Discrete vs. Continuous



Breaking Down Problems: State Space Search

Well-defined problems require six components:

- Initial State & Possible Actions
- Transition Model & Goal Test
- Path Cost Function & Solution



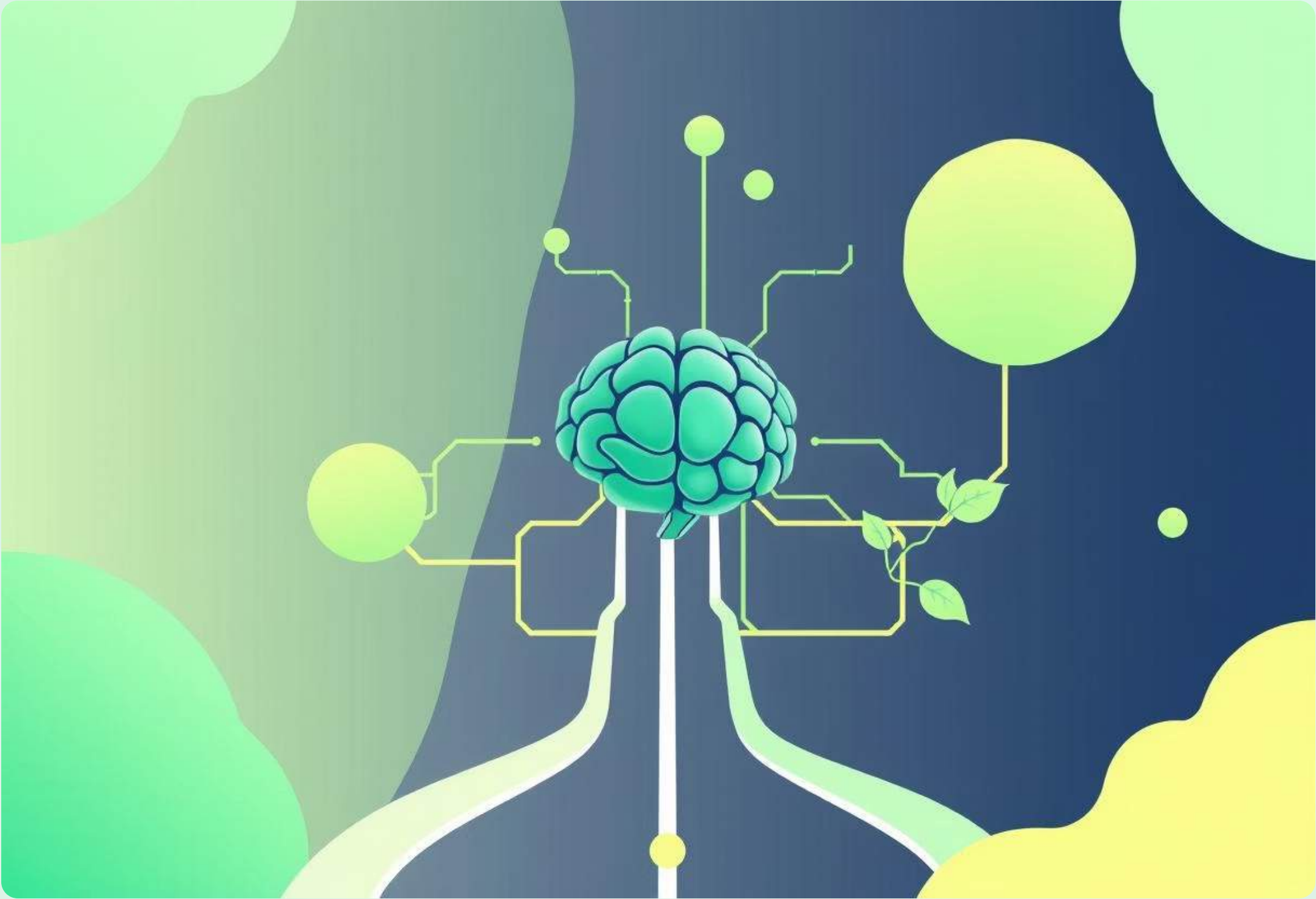
Optimizing the Search: Informed Strategies

1	2	3
Uninformed Search BFS, DFS, Iterative Deepening for completeness and memory efficiency, without domain knowledge.	Heuristics Intelligent guesswork (e.g., Manhattan distance) to guide search and achieve dramatic improvements.	Advanced Algorithms A* guarantees optimality; Hill Climbing for optimization, managing local maximum pitfalls.

Core Approaches to AI: Different Paths to Intelligence

Over decades, AI research has explored various paradigms, each offering distinct methods for systems to perceive, reason, and act. These approaches shape how we design intelligent agents and solve complex problems.

		
Symbolic AI Focuses on explicit knowledge representation and logical reasoning, often using rules and symbols to mimic human thought processes. • Rule-based systems • Expert systems • Logic programming	Machine Learning Enables systems to learn patterns from data without explicit programming, adapting and improving performance through experience. • Supervised/Unsupervised learning • Deep neural networks • Reinforcement learning	Probabilistic AI Deals with uncertainty and incomplete information by applying statistical methods and probability theory to make informed decisions. • Bayesian networks • Hidden Markov Models • Probabilistic graphical models



From Theory to Practice: Applying AI in Real-World Scenarios

Having explored core AI concepts, agent interactions, and various algorithmic approaches, it's crucial to understand how these theoretical foundations translate into practical applications that deliver tangible value.



Robotics & Automation

AI powers intelligent robots in manufacturing, healthcare, and exploration, enabling complex task execution and adaptive behavior in dynamic environments.

- Automated assembly lines
- Surgical robots
- Autonomous vehicles



Data Analysis & Prediction

Machine learning algorithms are at the heart of extracting insights from vast datasets, forecasting trends, and making data-driven decisions across industries.

- Financial market prediction
- Customer behavior analysis
- Medical diagnostics



Natural Language Processing

Enables computers to understand, interpret, and generate human language, revolutionizing communication and information access.

- Voice assistants
- Sentiment analysis
- Machine translation



Heuristic Value: Guiding the Search

Heuristic functions provide an "informed guess" about the cost or distance to a goal state, acting as a crucial guide for AI agents navigating complex problem spaces. They enable more efficient problem-solving by prioritizing promising paths and making intelligent decisions.

Cost Estimation

Heuristics approximate the true cost or remaining distance to the goal, transforming exhaustive searches into more targeted evaluations.

Efficiency & Speed

By pruning less promising branches, heuristics drastically reduce the computational effort and time required, making complex problems tractable.

Optimality vs. Practicality

While not always guaranteeing the absolute optimal solution, heuristics often find "good enough" solutions much faster, critical for real-time applications and large state spaces.



The Anatomy of Informed Search Costs

In AI search algorithms, especially informed ones like A*, agents navigate problem spaces by evaluating different paths based on various cost components. Understanding these costs is crucial for efficient decision-making.



Path Cost (g)

The actual cumulative cost incurred from the initial starting node to the current node. It measures the "journey traveled" so far.



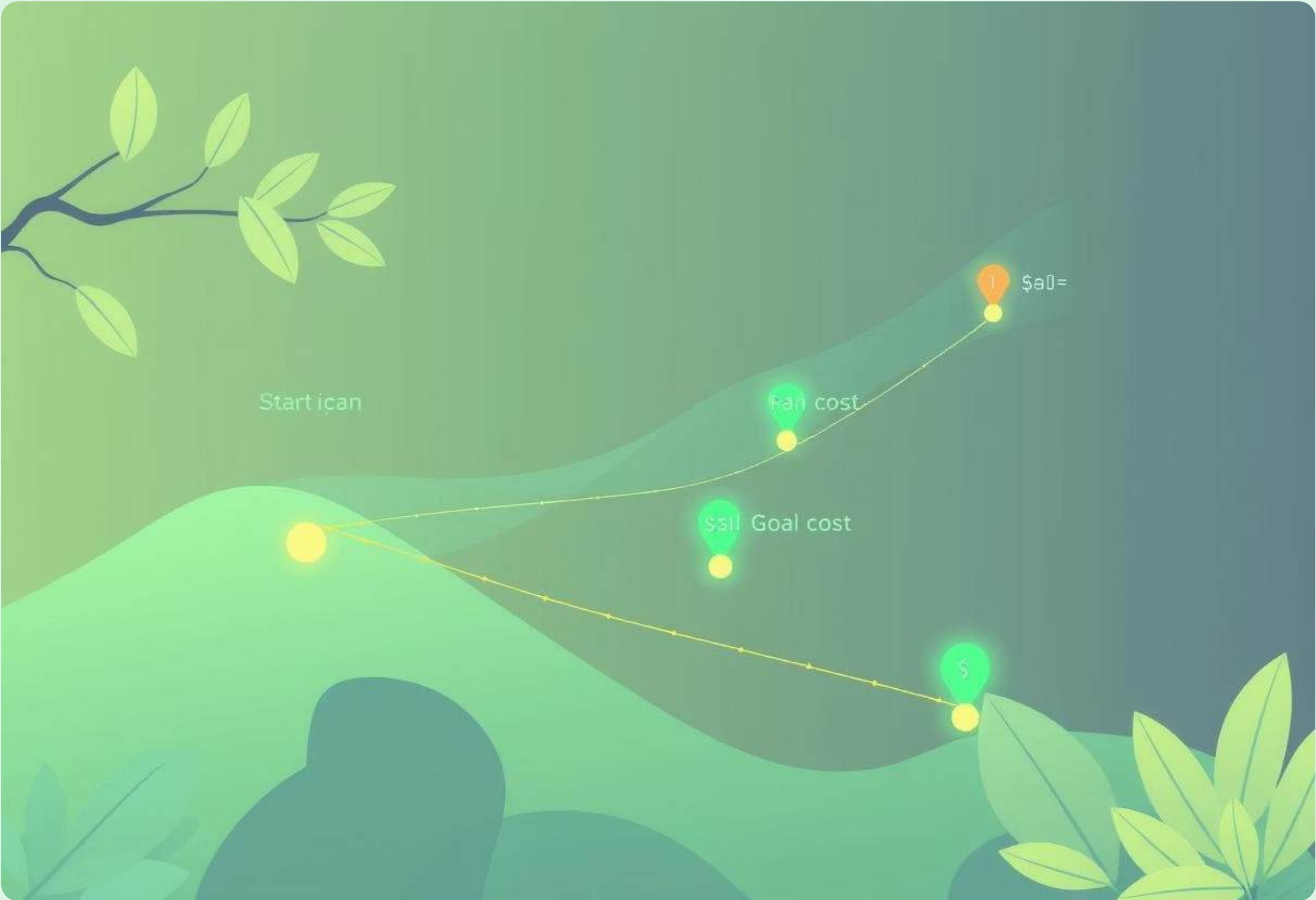
Goal Cost (h)

The estimated cost from the current node to the goal node. This is the heuristic's "informed guess" about remaining effort.



Total Estimated Cost (f)

The sum of path cost (g) and estimated goal cost (h), guiding the AI to the most promising next step:
 $f(n) = g(n) + h(n)$.





Conclusion: Beyond the Classroom

This 4-month journey transformed abstract AI concepts into working intelligence. This presentation itself demonstrates AI's capabilities, created using various AI tools for scripting, visualization, and narration.

1

Scripting & Refinement

ChatGPT for content structure and text refinement.

2

Visual Design

Canva AI templates and Miro/Lucidchart AI for diagrams.

3

Cinematic Visuals

Runway Gen-2 for immersive game sequences and animations.

4

Narration

ElevenLabs for professional voiceover and pacing.

This project highlights the synergy between academic theory and modern AI tools, creating a comprehensive and engaging learning experience.

Lab Tasks: Applying Search Algorithms in Practice

Understanding the theoretical underpinnings of search algorithms is one thing; applying them to solve practical problems in a laboratory or simulated environment reveals their strengths and weaknesses. Here's how different search paradigms tackle typical lab tasks:

Pathfinding & Mazes

Uninformed Search: Implementing BFS or DFS to find a path through a grid or maze, focusing on exploring all possibilities until a goal is found.

Optimal Route Planning

Informed Search: Developing A* or Greedy Best-First Search with heuristics to find the shortest or most efficient route on a weighted graph, like a road network.

Constraint Satisfaction

Local Search: Solving problems like the N-Queens puzzle or Sudoku by iteratively improving a solution, often using techniques like Hill Climbing or Simulated Annealing.

Strategic Game Play

Adversarial Search: Building agents that can play two-player games (e.g., Tic-Tac-Toe, Chess) using Minimax or Alpha-Beta pruning to anticipate opponent moves and optimize decisions.

These tasks provide concrete examples of how each algorithm's unique approach translates into tangible solutions, highlighting the trade-offs between completeness, optimality, and efficiency.

